A VERSION OF THE OLSSON-LASZLO BOUNDEDNESS LEMMA FOR A MORPHISM OF RINGED TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both assertions of the statement: the derived limit presentation, the placement of each truncated pushforward in the bounded below derived category and the degree comparison are all written out, with every cited input named.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans: the author's statement environment carries, inside a 'reference' environment at source lines 5797-5800, a provenance remark citing a published lemma of Olsson and Laszlo, and that remark is not part of the mathematical statement; it is therefore not delivered, and the counted statement is split into the two spans 5795-5796 and 5801-5823. No citation was supplied, invented or reconstructed, and no other text of the author was dropped.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The Olsson-Laszlo situation. Same chapter, Situation situation-olsson-laszlo. Supplies the hypotheses (3) and (4) refer to for both pairs (C, O, A) and (C', O', A').
- Unbounded complexes are derived limits of their truncations. Same chapter, Lemma lemma-olsson-laszlo. Writes K as the derived limit of its canonical truncations.
- The derived pushforward commutes with derived limits. Same chapter, Lemma lemma-Rf-commutes-with-Rlim. Moves the derived limit through Rf_*.
- The spectral sequence of a complex with respect to a functor. Derived Categories, Lemma derived-lemma-two-ss-complex-functor, together with Homology, Lemma homology-lemma-biregular-ss-converges. Shows that each truncated pushforward lies in the bounded below derived category of A'.

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- The modified Olsson-Laszlo comparison. Same chapter, Lemma lemma-olsson-laszlo-modified. Converts the comparison of truncations into the two displayed assertions.

Source reference. Original lemma 0D6S: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5707–5707. Navigation pointer invoked at proof line 5826; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0A07: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5150–5150. Navigation pointer invoked at proof line 5827; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 015J: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5742–5742. Navigation pointer invoked at proof line 5742; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 012W: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5838–5838. Navigation pointer invoked at proof line 5838; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D6T: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5726–5726. Navigation pointer invoked at proof line 5845; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Situation 0.1. Let $\mathcal{C}$ be a site. Let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}$. Let $\mathcal{A} \subset Mod(\mathcal{O})$ be a weak Serre subcategory. We assume the following is true: there exists a subset $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ such that

(1) every object of $\mathcal{C}$ has a covering whose members are in $\mathcal{B}$, and
(2) for every $V \in \mathcal{B}$ there exists an integer d_V and a cofinal system Cov_V of coverings of V such that

$$H^p(V_i, \mathcal{F}) = 0 \text{ for } \{V_i \to V\} \in \mathrm{Cov}_V,\ p > d_V,\ \text{and } \mathcal{F} \in \mathrm{Ob}(\mathcal{A})$$

Lemma 0.2. *Let $f : (Sh(\mathcal{C}), \mathcal{O}) \to (Sh(\mathcal{C}'), \mathcal{O}')$ be a morphism of ringed topoi. Let $\mathcal{A} \subset Mod(\mathcal{O})$ and $\mathcal{A}' \subset Mod(\mathcal{O}')$ be weak Serre subcategories. Assume there is an integer N such that*

(1) *$\mathcal{C}, \mathcal{O}, \mathcal{A}$ satisfy the assumption of Situation 0.1,*
(2) *$\mathcal{C}', \mathcal{O}', \mathcal{A}'$ satisfy the assumption of Situation 0.1,*
(3) *$R^p f_* \mathcal{F} \in \mathrm{Ob}(\mathcal{A}')$ for $p \geq 0$ and $\mathcal{F} \in \mathrm{Ob}(\mathcal{A})$,*
(4) *$R^p f_* \mathcal{F} = 0$ for $p > N$ and $\mathcal{F} \in \mathrm{Ob}(\mathcal{A})$,*

Then for K in $D_{\mathcal{A}}(\mathcal{O})$ we have

(a) *$Rf_* K$ is in $D_{\mathcal{A}'}(\mathcal{O}')$,*
(b) *the map $H^j(Rf_* K) \to H^j(Rf_*(\tau_{\geq -n} K))$ is an isomorphism for $j \geq N - n$.*

Proof. By Lemma 0D6S we have $K = R\lim \tau_{\geq -n}K$. By Lemma 0A07 we have $Rf_*K = R\lim Rf_*\tau_{\geq -n}K$. The complexes $Rf_*\tau_{\geq -n}K$ are bounded below. The spectral sequence

$$E_2^{p,q} = R^p f_* H^q(\tau_{\geq -n}K)$$

converging to $H^{p+q}(Rf_*\tau_{\geq -n}K)$ (Derived Categories, Lemma 015J) and assumption (3) show that $Rf_*\tau_{\geq -n}K$ lies in $D^+_{\mathcal{A}'}(\mathcal{O}')$, see Homology, Lemma 012W. Observe that for $m \geq n$ the map

$$Rf_*(\tau_{\geq -m}K) \longrightarrow Rf_*(\tau_{\geq -n}K)$$

induces an isomorphism on cohomology sheaves in degrees $j \geq -n + N$ by the spectral sequences above. Hence we may apply Lemma 0D6T to conclude. □